

PAGE 1 — PROJECT PLANNING PHASE

EDUGENIE

Project Planning Phase

Government Arts and Science College, Alangudi

College Code: BDU752

Department of Computer Science

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Academic Project Document

PAGE 2 — INTRODUCTION

1. INTRODUCTION

EduGenie is an educational support project designed to make learning more accessible, organized, and engaging for students. The project focuses on providing useful academic assistance through a simple and student-friendly digital platform.

Students often spend considerable time searching for learning materials, explanations, practice resources, and guidance. EduGenie is planned to bring these learning activities together in one platform. The system is intended to support students in understanding concepts, revising lessons, practicing questions, and improving their learning experience.

The Project Planning Phase defines what the EduGenie project is expected to achieve and how the development work will be organized. It identifies the major requirements, project scope, resources, activities, schedule, risks, and expected outcomes.

The planning process is important because it provides a clear direction before implementation begins. It also helps the project team divide responsibilities, estimate the required resources, and monitor progress throughout development.

Project Vision

The vision of EduGenie is to develop a simple, useful, and accessible educational platform that supports students in their academic learning and encourages independent study.

Project Mission

The mission is to provide organized educational assistance through an easy-to-use system while reducing the difficulty of finding appropriate learning resources.

PAGE 3 — PROJECT OVERVIEW

2. PROJECT OVERVIEW

Project Name: EduGenie

Domain: Education / Learning Support

Department: Computer Science

Institution: Government Arts and Science College, Alangudi

College Code: BDU752

2.1 Project Description

EduGenie is planned as a digital learning support system for students. The application can provide educational content and academic assistance in an organized manner. The system is designed with the needs of students in mind, with emphasis on simplicity, accessibility, and ease of use.

2.2 Proposed Major Features

- Student-friendly user interface
- Subject and topic-based learning support
- Educational notes and learning materials
- Question and answer support
- Practice and revision activities
- Search and navigation facilities
- Organized academic resources
- Simple and responsive design
- Basic user management, where required
- Progress or learning-support features, depending on implementation

2.3 Target Users

The primary users are students who need quick access to academic resources and learning assistance. Teachers or administrators may also be considered as supporting users for managing educational content, depending on the final system design.

2.4 Project Approach

The project will follow a systematic development approach. The team will first identify requirements, design the system, develop the required modules, test the application, correct identified issues, and finally prepare the documentation and demonstration.

PAGE 4 — PROBLEM STATEMENT

3. PROBLEM STATEMENT

Students use multiple sources to find study materials, explanations, practice questions, and other academic resources. When information is distributed across different websites, applications, books, and files, students may find it difficult to locate the required content quickly.

The absence of a simple, organized learning-support platform can create several challenges:

- Difficulty in finding relevant study materials.
- Time spent searching through multiple sources.
- Lack of organized topic-wise academic resources.
- Difficulty in revising concepts in a structured manner.
- Limited access to quick academic assistance.
- Students may not know which learning resource is appropriate for a particular topic.
- Repeated searching can reduce the efficiency of study time.

EduGenie is planned to address these challenges by organizing useful educational support in one accessible platform.

3.1 Need for the Project

A student-focused educational system can help learners access academic information in a structured way. By combining learning resources and support features, EduGenie can provide a convenient environment for study and revision.

3.2 Proposed Solution

The proposed solution is to develop EduGenie as a centralized educational support application. The system will organize relevant learning features and resources so that students can access them through a simple interface.

The project will concentrate on usability, clear navigation, organized content, and reliable operation. The final features will depend on the technologies and resources selected during development.

PAGE 5 — OBJECTIVES

4. OBJECTIVES

The main objective of EduGenie is to develop a student-friendly educational support system that makes academic learning more organized and accessible.

4.1 Primary Objectives

1. To provide a simple platform for students to access learning support.
2. To organize educational materials according to subjects and topics.
3. To reduce the time required to search for academic resources.
4. To support students during learning and revision.
5. To provide a clear and easy-to-use interface.
6. To improve accessibility to educational content.
7. To provide useful academic assistance through planned system features.
8. To create a scalable foundation for adding more educational features in the future.

4.2 Functional Objectives

- Allow users to navigate through available educational content.
- Provide subject/topic-based learning resources.
- Support question-and-answer or academic assistance features.
- Provide practice or revision facilities where implemented.
- Enable organized management of content.
- Provide appropriate feedback when users interact with the system.

4.3 Quality Objectives

The system should be:

- Usable — easy for students to understand and operate.
- Reliable — functions consistently under normal usage.
- Maintainable — easy to update and improve.
- Secure — user and system data should be handled appropriately.
- Responsive — pages and features should work efficiently.
- Scalable — future educational modules can be added when required.

PAGE 6 — SCOPE AND REQUIREMENTS

5. SCOPE OF THE PROJECT

The scope of EduGenie covers the planning and development of an educational support platform for students. The project focuses on academic assistance, organized learning resources, and a simple user experience.

5.1 In-Scope Activities

- Requirement analysis
- User interface design
- Educational content organization
- Development of core application modules
- Database or data management, where required
- Testing and debugging
- Documentation
- Final demonstration

5.2 Out-of-Scope Activities

The initial project may not include advanced features such as large-scale institutional administration, professional certification, complex payment systems, or highly specialized enterprise-level analytics unless they are specifically added during development.

6. PROJECT REQUIREMENTS

6.1 Hardware Requirements

- Computer or laptop for development
- Minimum 4 GB RAM recommended
- Adequate storage for development tools and project files
- Internet connection when online resources or services are required

6.2 Software Requirements

- Operating system such as Windows or Linux
- Code editor / IDE
- Appropriate programming language and development framework
- Database system, if required
- Web browser for testing
- Version control or project backup tools, where available

6.3 Human Resources

The project requires student developers who can perform requirement analysis, design, coding, testing, documentation, and presentation. Team members can divide responsibilities according to their skills.

PAGE 7 — PROJECT SCHEDULE AND RISK PLANNING

7. PROJECT SCHEDULE AND PLANNING

The project will be completed through a sequence of planned activities.

Phase 1 — Requirement Analysis

Identify user needs, project objectives, features, and system requirements.

Phase 2 — System Design

Prepare the application structure, user interface plan, database design where required, and module design.

Phase 3 — Development

Implement the planned modules and integrate the major features of EduGenie.

Phase 4 — Testing

Test individual modules and the complete application. Identify errors and correct them.

Phase 5 — Documentation

Prepare project reports, screenshots, diagrams, testing records, and other required documentation.

Phase 6 — Final Demonstration

Present the completed project and explain its objectives, features, implementation, and results.

7.1 Team Responsibility Planning

- Team Leader — coordination, task monitoring, integration support.
- Development Members — coding and module implementation.
- Design/Documentation Members — interface design, report preparation, and presentation.
- Testing Support — test cases, error identification, and verification.

Responsibilities may be shared among team members according to the project requirements.

7.2 Risk Management

Possible risks include:

- Delay in completing development tasks.
- Technical errors during implementation.
- Problems with data or database integration.
- Lack of sufficient testing time.
- Changes in project requirements.
- Loss of project files.

Risk reduction measures include maintaining backups, dividing work into smaller tasks, testing modules regularly, documenting progress, and communicating within the team.

7.3 Monitoring

Project progress will be monitored by checking completed tasks, pending work, identified issues, and upcoming activities. Regular team discussions can help ensure that the project remains on schedule.

PAGE 8 — EXPECTED OUTCOME AND CONCLUSION

8. EXPECTED OUTCOME

At the end of the project, EduGenie is expected to provide a functional educational support system with an organized and student-friendly interface.

Expected outcomes include:

- A working prototype/application of EduGenie.
- Organized access to educational resources.
- Easy navigation for students.
- Academic support through planned features.
- Improved convenience during study and revision.
- Properly tested and documented project modules.
- A foundation for future improvements and additional educational features.

8.1 FUTURE ENHANCEMENTS

Future versions of EduGenie may include additional subjects, improved personalization, advanced search, progress tracking, interactive quizzes, recommendation features, multilingual support, mobile application support, and other educational tools.

9. CONCLUSION

The Project Planning Phase establishes a clear foundation for the development of EduGenie. It defines the purpose, problem, objectives, scope, requirements, schedule, responsibilities, risks, and expected outcomes of the project.

Through systematic planning, the project team can organize development activities and use available resources effectively. The planned EduGenie system aims to provide students with a convenient and organized learning-support environment.

The next stages of the project will focus on system design, implementation, testing, documentation, and final presentation. Proper execution of the planned activities will help the team develop EduGenie into a useful academic project.

END OF PROJECT PLANNING PHASE